MAIS PAC

IPhO 02.10: Waves, Sound and the Doppler Effect · Lecture + Problems

# Waves, Sound and the Doppler Effect

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

**Contents & learning path**

1. What a wave is
2. Superposition, interference and standing waves
3. Sound: intensity, level and the decibel
4. The Doppler effect
5. Resonance in pipes and strings
6. Problems: reason, calculate, check
7. Further practice and provenance
8. Where to learn more

## 1. What a wave is

A wave is a disturbance that carries energy and momentum from one place to another **without carrying matter**. The medium — a string, water, air, or the electromagnetic field — oscillates about its own equilibrium while the pattern travels.

Two speeds are easy to confuse:

- The **wave speed** v: how fast a crest moves. It is set by the medium (tension and mass density for a string, bulk modulus and density for a fluid).
- The **particle speed**: how fast a piece of the medium moves. It is v_({particle}) = ω A at most, and it reverses every half cycle.

For a sinusoidal wave,

$$y(x,t) = A\sin(kx - \omega t),\ k = \frac{2\pi}{\lambda},\ \omega = 2\pi f,\ v = f\lambda = \frac{\omega}{k}. \quad (1)$$

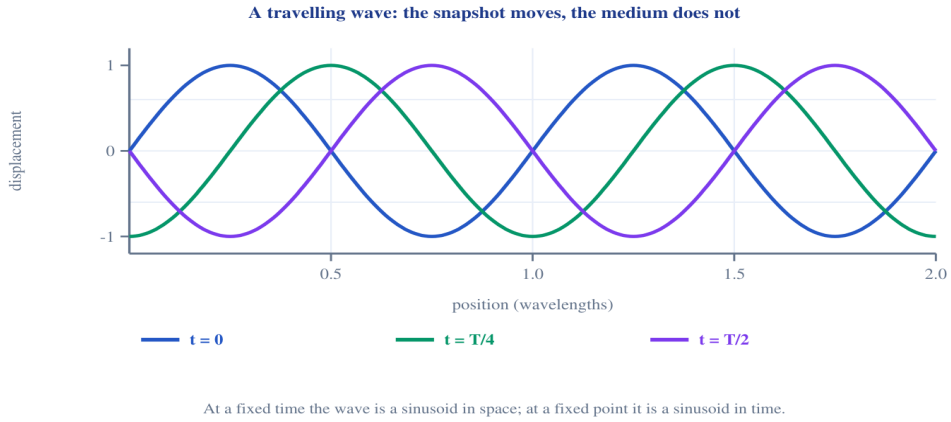



**Figure.** At a fixed time the wave is a sinusoid in space; at a fixed point it is a sinusoid in time.

Two numbers carry the whole wave: λ tells you how the pattern is laid out in space, f how it is laid out in time, and v links them. In a given medium v is fixed by the material, so a higher frequency means a shorter wavelength.

Important special cases:

- **String:** v = √(T/μ) with T the tension and μ the mass per unit length.
- **Sound in a gas:** v = √(γ p/ρ) = √(γ RT/M), which is why sound travels faster in warm air, faster in helium, and not at all in vacuum.
- **Light:** v = c/n in a medium of refractive index n; the frequency is unchanged when light enters glass, so the wavelength shortens.

## 2. Superposition, interference and standing waves

Waves obey the **principle of superposition**: at any point the displacement is the sum of the displacements of the individual waves. That is not obvious — it follows from the fact that the wave equation is linear — and it is the source of almost all of wave optics and acoustics.

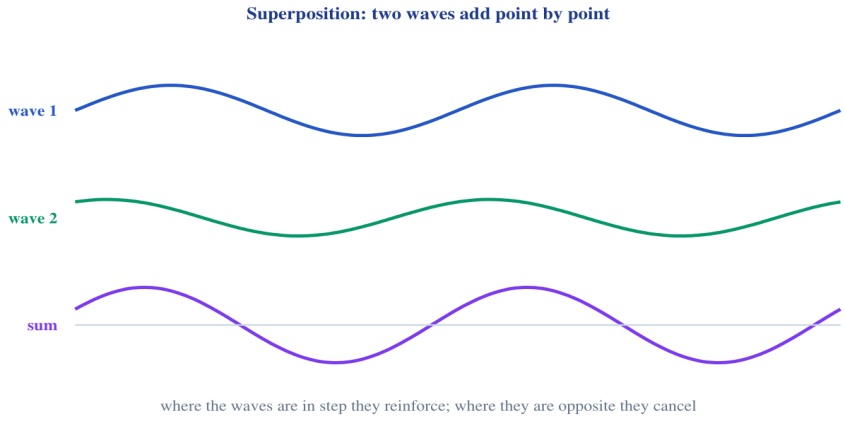



**Figure 2.** The sum is still a sinusoid of the same frequency, with an amplitude between the difference and the sum of the two. Only the phase and the amplitudes decide which.

- Two waves of the same frequency add to a single wave of that frequency. If they are in phase the amplitudes add (**constructive**); if opposite, they subtract (**destructive**).
- Two waves of slightly different frequency produce **beats**: the amplitude rises and falls at the difference frequency, f_({beat}) = |f_1 - f_2|.

- A wave meeting its own reflection produces a **standing wave**: fixed nodes where the two always cancel, and antinodes where they always reinforce.

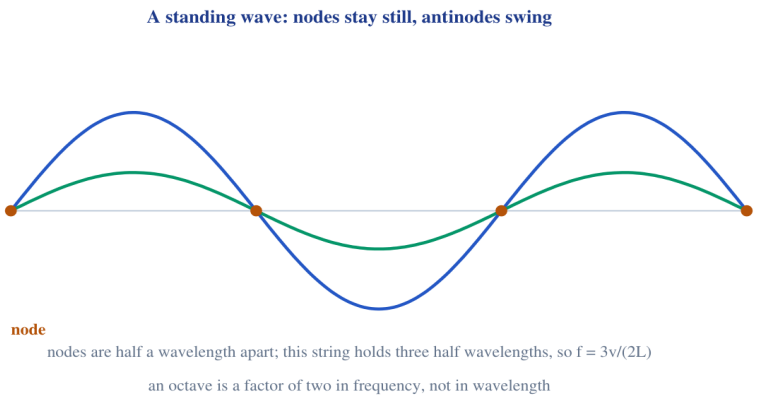



**Figure 3.** A standing wave is not a wave travelling nowhere: it is two travelling waves of equal amplitude moving in opposite directions, and only the frequencies that fit the boundary conditions survive.

On a string of length L fixed at both ends, the ends must be nodes, so only wavelengths with L = nλ/2 fit:

$$f_n = (nv)/(2L), n = 1, 2, 3, \quad (2)$$

The n=1 mode is the fundamental, and the higher modes are the harmonics — the whole basis of musical instruments.

## 3. Sound: intensity, level and the decibel

Sound is a longitudinal pressure wave. Its intensity is power per unit area,

$$I = (P)/(4\pi r^2) \{(a point source radiating in all directions)\}, \quad (3)$$

so intensity falls as the inverse square of the distance. Because the ear responds logarithmically, intensity is usually quoted as a level:

$$L = 10log _{10}(I)/(I_0) \{dB\}, I_0 = 10^(-12) \{W,m^(-2)\}. \quad (4)$$

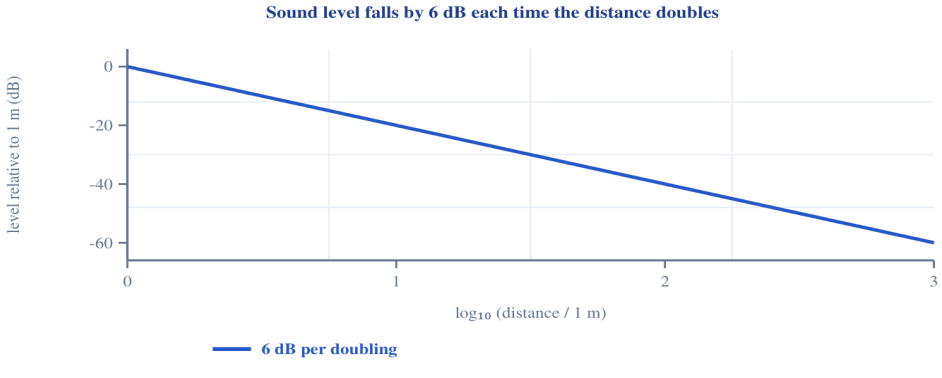



**Figure.** The intensity itself falls as 1/r², which is a straight line once it is expressed in decibels.

The rules worth memorising: a factor of 10 in intensity is 10 {dB}; a factor of 2 is about 3 {dB}; a factor of 4 in distance is -12 {dB}. Adding a second identical source adds 3 {dB}, not 6.

## 4. The Doppler effect

If the source or the observer moves, the **observed frequency** differs from the emitted one. The reason is simple geometry: the crests emitted while the source approaches are packed closer together, so they arrive more often.

$$f_{\{obs\}} = f_{\{source\}},(v \pm v_{\{obs\}})/(v \mp v_{\{source\}}). \quad (5)$$

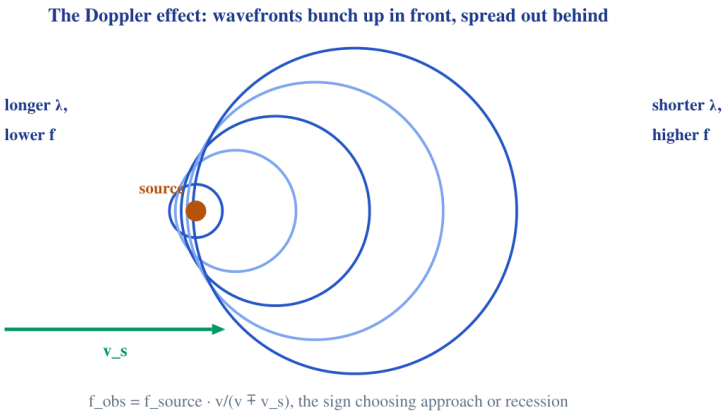



**Figure 4.** The frequency does not change at the source: what changes is how many crests per second reach the observer. The same rule gives the redshift of receding galaxies and the change of pitch of an ambulance siren.

The sign convention is the only thing to get right: use the upper signs when the two are approaching (higher frequency), the lower signs when they are receding (lower pitch, like the drop in pitch of a passing ambulance).

Note the asymmetry in sound: moving the source and moving the observer give **different** shifts, because sound travels through a medium that defines a preferred frame. For light there is no medium, the two cases are symmetric, and the correct formula is relativistic:

$$f_{\{obs\}} = f_{\{source\}}\sqrt{((1 \pm \beta)/(1 \mp \beta))}, \beta = (v)/(c). \quad (6)$$

For $v \ll c$ this reduces to $\Delta f/f \approx v/c$ — the **redshift** of receding galaxies and the **blueshift** of approaching stars.

**Interactive experiment: Doppler shift for a moving source and observer — open the online handout to use it.**

## 5. Resonance in pipes and strings

A pipe closed at one end has a node at the closed end and an antinode at the open end, so it fits an odd number of quarter wavelengths:

$$L = ((2n-1)\lambda)/(4)\ f_n = ((2n-1)v)/(4L),\ n = 1, 2, 3, \quad (7)$$

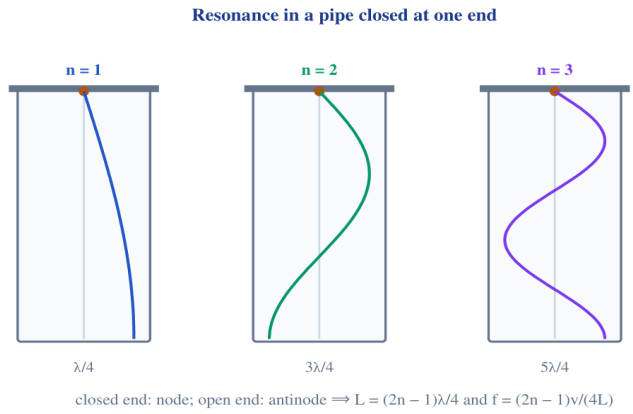



**Figure 6.** A closed end must be a node and an open end an antinode, so the pipe holds an odd number of quarter wavelengths — the odd harmonics only. A pipe open at both ends holds any number of half wavelengths, which is why it sounds brighter.

A pipe open at both ends has antinodes at both ends and fits any number of half wavelengths, f_n = nv/(2L): it has all the harmonics, which is why an open flute sounds brighter than a stopped clarinet.

The **end correction** matters in real instruments: the air just outside an open end also vibrates, so the effective length is about L + 0.6r for a pipe of radius r. In an exam, if the numbers do not quite fit, an end correction is the first thing to suspect.

**The waves checklist.** Which quantity is fixed by the medium? (usually v) → use v = fλ to convert between space and time → superposition for anything with two waves (interference, beats, standing waves) → boundary conditions decide which wavelengths fit (node at a fixed end, antinode at an open end) → for a moving source, work out whether the crest spacing or the arrival rate changes, and use the Doppler formula with the correct signs → check the answer in the limits v_({source})→0 and v_({source})→ v.

## 6. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

**[1] Problem 1 — A wave on a string.** A string of mass per unit length 2.0 {g,m^(-1)} is stretched to a tension of 80 {N}. A wave of frequency 120 {Hz} travels along it. (a) Find the wave speed. (b) Find the wavelength. (c) Find the maximum transverse speed of a point on the string if the amplitude is 2.0 {mm}, and compare it with the wave speed.

Worked solution →

**[1] Problem 2 — Harmonics of a string.** A guitar string of length 0.65 {m} is tuned so that its fundamental is 196 {Hz}. (a) Find the wave speed on the string. (b) Find the frequencies of the next three harmonics. (c) Explain what happens to the fundamental if the string is pressed at its midpoint, and how that relates to an octave.

Worked solution →

**[2] Problem 3 — Sound intensity and decibels.** A loudspeaker radiates 2.0 {W} of sound uniformly in all directions. (a) Find the intensity at 3.0 {m}. (b) Find the sound level there in dB. (c) Find the distance at which the level has fallen by 20 {dB}, and the intensity there.

Worked solution →

**[2] Problem 4 — Beats.** Two tuning forks of 440 {Hz} and 444 {Hz} sound together. (a) Find the beat frequency. (b) Explain what a listener hears, and what happens if a third fork of 442 {Hz} is added. (c) Explain how a piano tuner uses beats to tune a string to a reference.

Worked solution →

**[2] Problem 5 — A pipe closed at one end.** A pipe closed at one end resonates at 320 {Hz} and at 960 {Hz} but at no frequency in between. (a) Show that these are consistent with a closed pipe and find the fundamental frequency. (b) Find the length of the pipe, taking the speed of sound as 340 {m,s^(-1)}. (c) Find the next two resonant frequencies and explain why no resonance occurs at 640 {Hz}.

Worked solution →

**[2] Problem 6 — The Doppler effect for an ambulance.** An ambulance siren emits 700 {Hz} and the vehicle moves at 25 {m,s^(-1)} towards a stationary observer, then away. Take v = 340 {m,s^(-1)}. (a) Find the frequency heard while approaching. (b) Find the frequency heard after passing. (c) Find the fractional change across the pass, and explain why an observer hears a sudden drop rather than a gradual one.

Worked solution →

**[2] Problem 7 — Moving source and moving observer.** A source emits 500 {Hz} and moves at 30 {m,s^(-1)} towards an observer who is also moving at 20 {m,s^(-1)} towards the source. (a) Find the observed frequency. (b) Compare it with the case where the source is stationary and the observer moves at 50 {m,s^(-1)}. (c) Explain why the two answers differ, and what this says about the medium.

Worked solution →

**[3] Problem 8 — Interference from two speakers.** Two loudspeakers 3.0 {m} apart emit 200 {Hz} in phase. A listener walks along a line parallel to the speakers at 4.0 {m} in front of them. (a) Find the wavelength. (b) Find the path difference when the listener is directly in front of one speaker, and decide whether it is a maximum or a minimum. (c) Find how many maxima the listener crosses walking from one side to the other.

Worked solution →

**[3] Problem 9 — Standing waves and resonance in a tube.** A tube of length 0.50 {m}, open at both ends, is driven by a small source at one end. (a) Find its three lowest resonant frequencies, with v = 340 {m,s^(-1)}. (b) Find the frequencies if one end is closed. (c) Explain why a closed tube can sound an octave lower than an open tube of the same length.

Worked solution →

**[3] Problem 10 — A wave on a string with a boundary.** A pulse travels along a light string and meets a heavier string tied to it. (a) Describe qualitatively what happens: what is transmitted and what is reflected, and with what phase. (b) Explain what changes if the second string is lighter than the first. (c) Explain how this is related to the reflection of a wave at a fixed end and at a free end.

Worked solution →

## Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

**Sources and scope.** Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

## Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

MIT OpenCourseWare 8.03, Vibrations and Waves

Oscillators, resonance, superposition and optics with demonstrations.